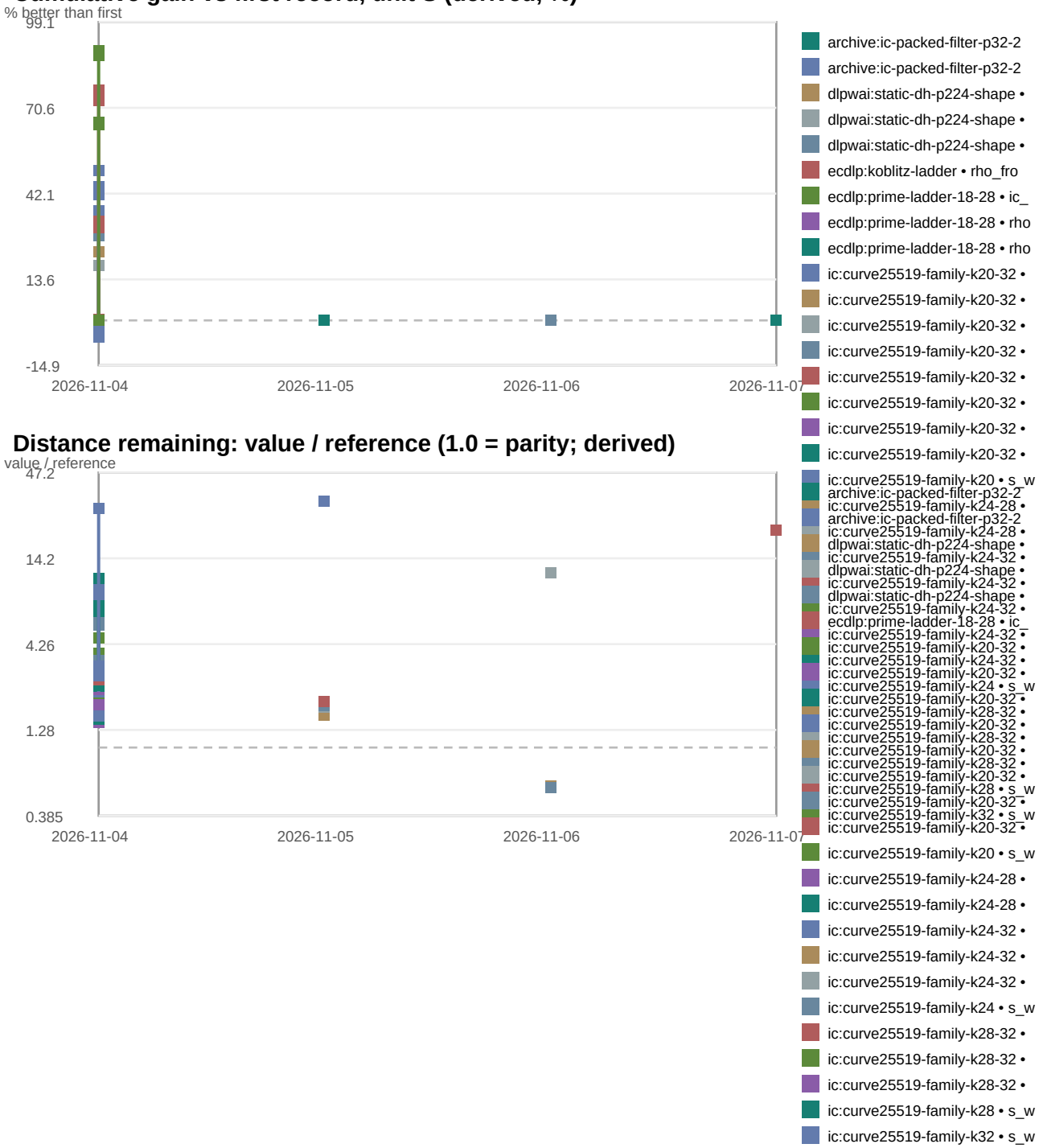
# Benchmark progress: crypto

From bench.record/v1 records in Git; raw data in S3 by SHA-256. Derived numbers are labelled derived.

Records: open 0, measured 53, invalid 1, censored 4. Headline unit S: group operations / sqrt(n); the ECDLP cost unit, every phase priced.

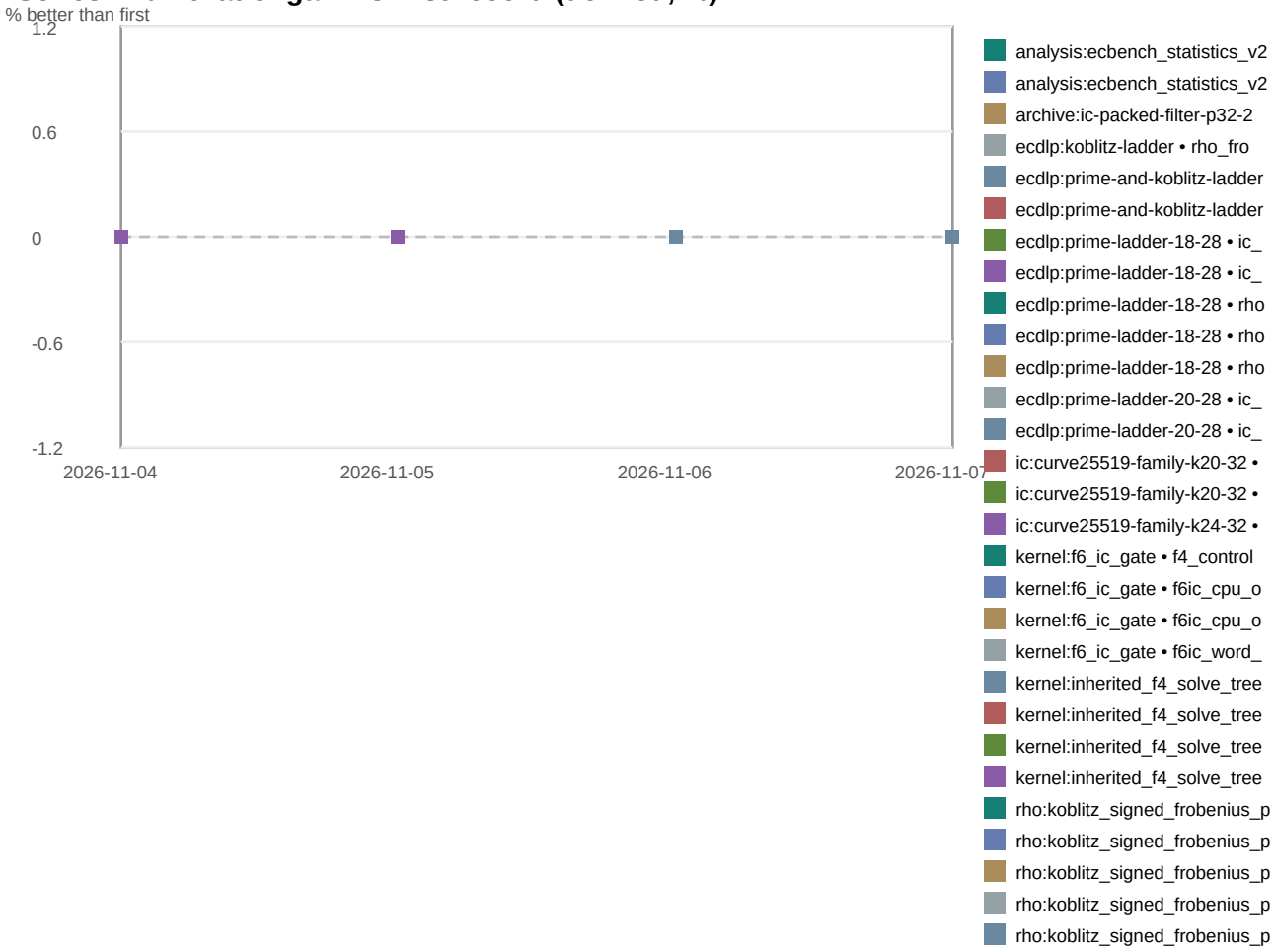
## Cumulative gain vs first record, unit S (derived, %)




## Distance remaining: value / reference (1.0 = parity; derived)

## Series in unit count: gain vs first record (derived, %)

## Series in unit ratio: gain vs first record (derived, %)



# Series summary

| problem • metric | unit | first | latest | gain vs first | value/ref | pts |
|---|---|---|---|---|---|---|
| analysis:ecbench_statistics_v2 • censored_ratio_ | ratio | 648.2 | 648.2 | ? | ? | 1 |
| analysis:ecbench_statistics_v2 • heavy_tail_rows | count | 71 | 71 | ? | ? | 1 |
| analysis:ecbench_statistics_v2 • ops_interval_wi | ratio | 13.1 | 13.1 | ? | ? | 1 |
| archive:ic-packed-filter-p32-20261029 • s_prime_ | S | 31.6936 | 31.6936 | 0% | 31.4718 | 1 |
| archive:ic-packed-filter-p32-20261029 • s_prime_ | S | 63.4961 | 63.4961 | 0% | 31.6255 | 1 |
| archive:ic-packed-filter-p32-20261029 • wall_per | ratio | 0.9601 | 0.9601 | 0% | ? | 1 |
| archive:icb-large-prime-n53-s1-20261029 • report | count | 240 | 240 | ? | ? | 1 |
| archive:icb-large-prime-n53-s2-20261029 • report | count | 240 | 240 | ? | ? | 1 |
| archive:icb-large-prime-n61-s1-20261029 • report | count | 72 | 72 | ? | ? | 1 |
| archive:icb-large-prime-n61-s2-20261029 • report | count | 60 | 60 | ? | ? | 1 |
| dlpwai:static-dh-p224-shape • s_w0_cheon_pminus1 | S | 0.5739 | 0.5739 | 0% | 0.5814 | 2 |
| dlpwai:static-dh-p224-shape • s_w0_cheon_pminus1 | S | 12.2426 | 12.2426 | 0% | 11.6066 | 2 |
| dlpwai:static-dh-p224-shape • s_w0_cheon_pplus1_ | S | 0.5819 | 0.5819 | 0% | 0.5746 | 2 |
| ecdlp:koblitz-ladder • rho_frob_v3_over_v2_ops | ratio | 0.459 | 0.459 | 0% | ? | 1 |
| ecdlp:koblitz-ladder • rho_frob_v3_s_k43 | S | 0.142 | 0.142 | 0% | ? | 1 |
| ecdlp:prime-and-koblitz-ladder • rho_frob_v4_ove | ratio | 0.9685 | 0.9685 | 0% | ? | 1 |
| ecdlp:prime-and-koblitz-ladder • rho_negation_v5 | ratio | 0.9718 | 0.9718 | 0% | ? | 1 |
| ecdlp:prime-ladder-18-28 • ic_auto_over_b32_ops | ratio | 0.2469 | 0.2469 | 0% | ? | 1 |
| ecdlp:prime-ladder-18-28 • ic_auto_over_rho_v3_o | ratio | 12.7982 | 12.7982 | 0% | ? | 1 |
| ecdlp:prime-ladder-18-28 • ic_auto_s_r28 | S | 19.188 | 19.188 | 0% | 20.9934 | 1 |
| ecdlp:prime-ladder-18-28 • rho_negation_v3_over_ | ratio | 0.8643 | 0.8643 | 0% | ? | 1 |
| ecdlp:prime-ladder-18-28 • rho_negation_v3_s_r24 | S | 0.923 | 0.923 | 0% | ? | 1 |
| ecdlp:prime-ladder-18-28 • rho_negation_v3_s_r28 | S | 0.923 | 0.923 | 0% | ? | 1 |
| ecdlp:prime-ladder-18-28 • rho_negation_v4_over_ | ratio | 0.9802 | 0.9802 | 0% | ? | 1 |
| ecdlp:prime-ladder-18-28 • rho_plain_v3_over_pla | ratio | 0.9391 | 0.9391 | 0% | ? | 1 |
| ecdlp:prime-ladder-20-28 • ic_auto_over_tuned_p2 | ratio | 0.998 | 0.998 | 0% | ? | 1 |
| ecdlp:prime-ladder-20-28 • ic_auto_over_tuned_p2 | ratio | 0.8848 | 0.8848 | 0% | ? | 1 |
| ic:curve25519-family-k20-32 • ratio_w0_k24 | ratio | 0.9375 | 0.9375 | 0% | ? | 1 |
| ic:curve25519-family-k20-32 • ratio_w0_k32 | ratio | 0.8802 | 0.8802 | 0% | ? | 1 |
| ic:curve25519-family-k20-32 • s_w0_incumbent_k20 | S | 2.5746 | 2.5746 | 0% | 1.7601 | 1 |
| ic:curve25519-family-k20-32 • s_w0_incumbent_k24 | S | 2.1638 | 1.6735 | 22.6592% | 1.429 | 2 |
| ic:curve25519-family-k20-32 • s_w0_incumbent_k28 | S | 1.9211 | 1.5717 | 18.1849% | 1.4943 | 2 |
| ic:curve25519-family-k20-32 • s_w0_incumbent_k32 | S | 2.11 | 1.5152 | 28.1881% | 1.5539 | 2 |
| ic:curve25519-family-k20-32 • s_w0_mean5curves_k | S | 2.601 | 2.601 | 0% | 1.5669 | 1 |
| ic:curve25519-family-k20-32 • s_w0_mean5curves_k | S | 2.111 | 2.111 | 0% | 1.7695 | 1 |
| ic:curve25519-family-k20-32 • s_w0_mean5curves_k | S | 1.869 | 1.869 | 0% | 1.7971 | 1 |
| ic:curve25519-family-k20-32 • s_w0_mean5curves_k | S | 1.93 | 1.93 | 0% | 1.9052 | 1 |
| ic:curve25519-family-k20 • s_w0_incumbent | S | 8.3478 | 4.8598 | 41.7827% | 3.2356 | 5 |
| ic:curve25519-family-k24-28 • s_w0_incumbent_k24 | S | 2.5842 | 2.5842 | 0% | 2.0304 | 1 |
| ic:curve25519-family-k24-28 • s_w0_incumbent_k28 | S | 2.3732 | 2.3732 | 0% | 2.3865 | 1 |
| ic:curve25519-family-k24-32 • best_train_s_w0_k2 | S | 2.713 | 2.713 | 0% | ? | 1 |
| ic:curve25519-family-k24-32 • best_train_s_w0_k3 | S | 2.768 | 2.768 | 0% | ? | 1 |
| ic:curve25519-family-k24-32 • ratio_w0_k32 | ratio | 0.9106 | 0.9106 | 0% | ? | 1 |
| ic:curve25519-family-k24-32 • s_w0_seedrun_k24 | S | 2.1055 | 2.1055 | 0% | 1.942 | 1 |
| ic:curve25519-family-k24-32 • s_w0_seedrun_k28 | S | 1.8835 | 1.8835 | 0% | 1.8154 | 1 |
| ic:curve25519-family-k24-32 • s_w0_seedrun_k32 | S | 1.8423 | 1.8423 | 0% | 1.8232 | 1 |
| ic:curve25519-family-k24 • s_w0_incumbent | S | 6.607 | 3.3182 | 49.7782% | 2.7163 | 5 |
| ic:curve25519-family-k28-32 • s_w0_incumbent | S | 2.5798 | 2.5798 | 0% | 2.5891 | 1 |
| ic:curve25519-family-k28-32 • s_w0_seed_mean_k28 | S | 1.969 | 1.969 | 0% | 1.8593 | 1 |
| ic:curve25519-family-k28-32 • s_w0_seed_mean_k32 | S | 1.781 | 1.781 | 0% | 1.8229 | 1 |
| ic:curve25519-family-k28 • s_w0_incumbent | S | 10.9395 | 2.576 | 76.452% | 2.7572 | 7 |
| ic:curve25519-family-k32 • s_w0_incumbent | S | 24.1377 | 2.519 | 89.5641% | 2.7453 | 5 |
| kernel:f6_ic_gate • f4_control_word_xors_off_ove | ratio | 1 | 1 | ? | ? | 1 |
| kernel:f6_ic_gate • f6ic_cpu_off_over_on_max | ratio | 10.29 | 10.29 | -0% | ? | 1 |
| kernel:f6_ic_gate • f6ic_cpu_off_over_on_min | ratio | 2.67 | 2.67 | -0% | ? | 1 |

| | | | | | | |
|---|---|---|---|---|---|---|
| kernel:f6_ic_gate • f6ic_word_xors_off_over_on_m | ratio | 1.84 | 1.84 | -0% | ? | 1 |
| kernel:inherited_f4_solve_tree • wall_seq_over_p | ratio | 0.936 | 0.936 | -0% | ? | 1 |
| kernel:inherited_f4_solve_tree • wall_seq_over_p | ratio | 2.883 | 2.883 | -0% | ? | 1 |
| kernel:inherited_f4_solve_tree • wall_seq_over_p | ratio | 1.589 | 1.589 | -0% | ? | 1 |
| kernel:inherited_f4_solve_tree • work_par_over_s | ratio | 2.865 | 2.865 | 0% | ? | 1 |
| registry-rebinding:ic-aimed-gather • raw_pins_re | count | 211 | 211 | ? | ? | 1 |
| rho:koblitz_signed_frobenius_panel • s_legacy_ov | ratio | 0.718 | 0.718 | -0% | ? | 1 |
| rho:koblitz_signed_frobenius_panel • s_legacy_ov | ratio | 1.0948 | 1.0948 | -0% | ? | 1 |
| rho:koblitz_signed_frobenius_panel • s_legacy_ov | ratio | 1.4723 | 1.4723 | -0% | ? | 1 |
| rho:koblitz_signed_frobenius_panel • s_legacy_ov | ratio | 1.0653 | 1.0653 | -0% | ? | 1 |
| rho:koblitz_signed_frobenius_panel • s_legacy_ov | ratio | 1.0008 | 1.0008 | -0% | ? | 1 |

Every value is quoted from the record it names, in that record’s unit. Percentages and ratios under ’derived’ are computed here from those values and are not measurements. Series in different units are never compared.